



SPACE DESIGN LOAD SUMMARY

Kitchen_4

2026-09-27

Level: Level 0 · Internal condition: 1 Bed Apt. Kitchen - Kitchen_4

DESIGN CRITERIA

	Heating	Cooling
Room set point	-50.0 °C	150.0 °C
Outdoor dry bulb	—	—
Outdoor RH (coincident)	—	—
ROOM HUMIDITY		
Humidification set point	—	
lower RH limit		
Dehumidification set point	—	
upper RH limit		

SIZING (TAS DESIGN LOADS)

	Heating	Cooling
Design load	0 W	0 W
Design load per area	0.0 W/m ²	0.0 W/m ²
Sizing multiplier	not set	not set
Design loads: from Tas sizing — date/currency not recorded		

HEATING

Design day	Leeds_TRY ANN HTG 100% CONDS DB	
	Design day	Full year
Peak sensible load	1,706 W	911 W
Peak hour	13:00–14:00	1 Jan 00:00–01:00 (HOY 1)
Room dry bulb	14.4 °C	13.6 °C
Room resultant	13.0 °C	12.8 °C
Room RH	22 %	58 %
Room humidity ratio	2.2 g/kg	5.6 g/kg
Outdoor dry bulb	—	3.3 °C
Outdoor RH	—	100 %

SENSIBLE LOAD COMPONENTS AT PEAK

Component	Design day (W)	Full year (W)
Equipment (sensible)	0	50
Infiltration / ventilation	-311	-157
Building heat transfer	-686	-478
External conduction — opaque	-481	-215
External conduction — glazing	-228	-111

Zero at both peaks (not listed): Solar, Lighting, Occupancy (sensible), Air movement (inter-zone), Air handling unit, Occupancy (latent), Equipment (latent)
Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

COOLING

Design day	Leeds_TRY ANN CLG 0% CONDS DB=>GRad	
	Design day	Full year
Peak sensible load	1,288 W	863 W
Peak hour	00:00–01:00	19 Jul 19:00–20:00 (HOY 4796)
Room dry bulb	20.1 °C	19.1 °C
Room resultant	21.3 °C	19.9 °C
Room RH	88 %	58 %
Room humidity ratio	13.0 g/kg	7.9 g/kg
Outdoor dry bulb	—	15.4 °C
Outdoor RH	—	80 %

SENSIBLE LOAD COMPONENTS AT PEAK

Component	Design day (W)	Full year (W)
Solar	0	61
Lighting	0	150
Occupancy (sensible)	0	19
Equipment (sensible)	50	300
Infiltration / ventilation	-13	-137
Building heat transfer	851	374
External conduction — opaque	442	153
External conduction — glazing	-42	-56

LATENT COMPONENTS AT PEAK

Component	Design day (W)	Full year (W)
Occupancy (latent)	0	14

Zero at both peaks (not listed): Air movement (inter-zone), Air handling unit, Equipment (latent)
Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

RESULTS

	Heating	Cooling
Status	Available	Available
Result source	SAM.Analytical.Tas	SAM.Analytical.Tas
Read into model	27 Sep 2026 20:42	27 Sep 2026 20:42
Matches current model	not recorded	not recorded

Peak hour: a design-day peak has an hour of the day only, no date; a full-year peak shows day and hour, no year, and its hour of the year (HOY 1 = 1 Jan 00:00–01:00, HOY 8760 = 31 Dec 23:00–24:00).
Design loads (sizing) and simulated peaks are shown side by side for information; no acceptance rule is applied.